DE LA SALLE HOLY CROSS COLLEGE HIGH

Answer Booklet [50]

Name:

SECTION A

Question 1 [10]

1. Four possible answers are given for each of the following questions. Select the most correct answer and write the letter corresponding to the question in the table below.

Question number	1.1.	1.2.	1.3.	1.4.	1.5.
Answer					

1.1. Which of the following processes takes place during cellular respiration?
A. Splitting of the water molecules
B. Liberation of oxygen
C. ADP is changed into ATP
D. CO_2 is taken in (2)

1.2. The smallest, most specific category of classification is:
A. Phylum
B. Species
C. Kingdom
D. Order (2)

1.3. The primary function of the cell nucleus is:
A. Storage of nutrients.
B. Respiration.
C. Cell functioning.
D. Reproduction. (2)

1.4. The small projections found on a plant root that help with the uptake of nutrients is called:
A. Hair-like follicles
B. Root hairs
C. Root cells
D. Projectiles (2)

1.5. Facilitated diffusion through a cell membrane:
A. requires energy and goes against the concentration gradient.
B. is the movement of small molecules into and out of the cell through the pores between phospholipids.
C. does not require energy and involves the net movement of molecules from a high to a low concentration.
D. does not involve protein carriers. (2)

Question 2 **[10]**

2. Write the correct biological term for the following definitions:

2.1. The phase of mitosis where the chromosomes are pulled toward the cell poles. (2)

2.2. The process our body carries out to make energy. (2)

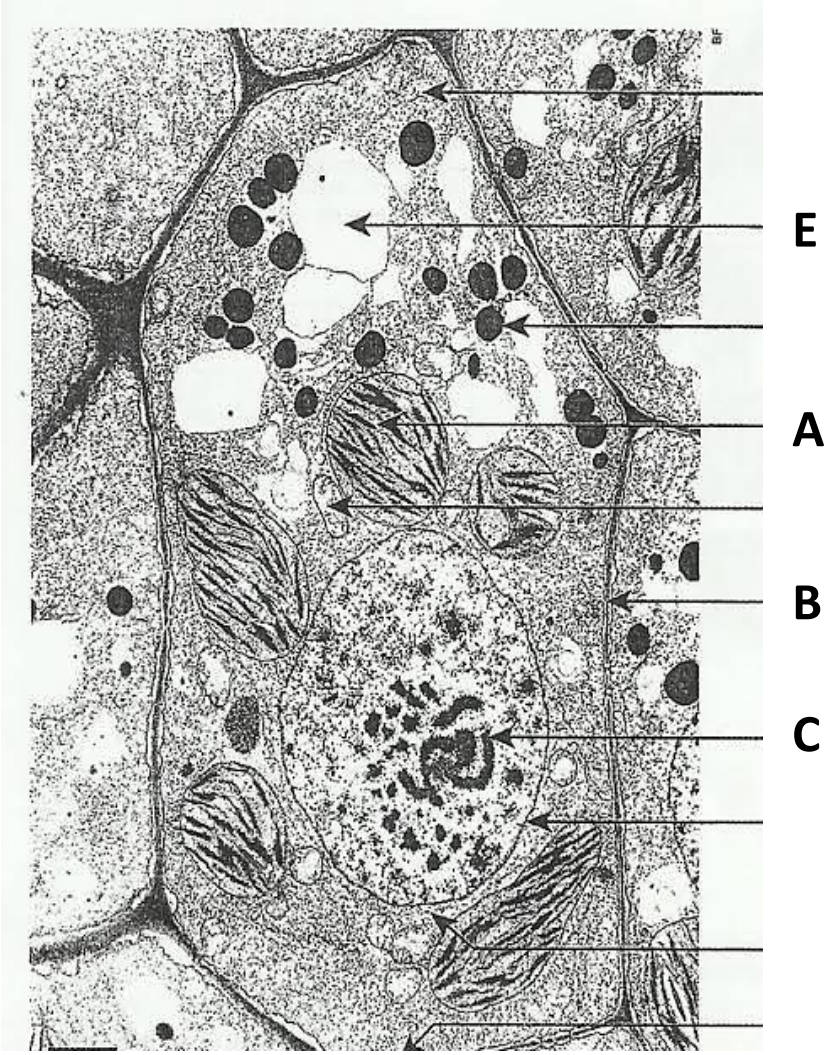
2.3. A stack of thylakoids. (2)

2.4. The muscular contraction of the oesophagus and small and large intestine. (2)

2.5. A visual representation of a one-way path of energy exchange through feeding between organisms. (2)

Question 3 **[10]**

3. Study the following diagram and answer the short questions that follow. Include all relevant information.



Adapted from: https://quizlet.com/219493626/aice-biology-chapter-1cell-electron-micrograph-labeling-diagram/

3.1. This diagram is a representation of: (2)

3.2. Label E is showing: (2)

3.3. What is the function of label C? (2)

3.4. Label B allows all substances into a cell. True or false? (2)

3.5. Label A represents: (2)

Question 4 **[10]**

4. Match the defintions in column B with the words in column A. Write only the correct letter next to the word.

Column A	Correct letter	Column B
4.1. Habitat		A. The area or environment where an organism or ecological community normally lives or occurs.
4.2. Biotic		B. Liquid content of the mitochondrion.
4.3. Niche		C. The passive movement of molecules or particles along a concentration gradient, or from regions of higher to regions of lower concentration.
4.4. Stroma		D. The ecological role and space that an organism fills in an ecosystem.
4.5. Diffusion		E. All plants, animals, fungi, protists and bacteria within an ecosystem.
		F. A major constituent of the fossil record made of Cyanobacteria as well as other prokaryotes.
		G. Liquid content of the chloroplast.

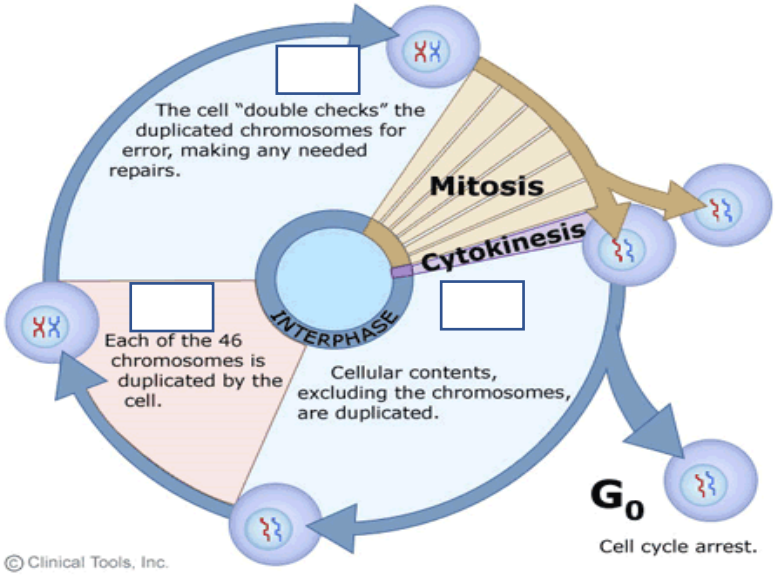
Question 5 **[10]**

5. The following diagram of the cell cycle has been provided. Add the following labels and answer the questions that follow. (You may include the label on either the right or left hand side of the diagram for clarity.

   Growth phase 1

   Growth phase 2

   Synthesis (3)



Adapted from: https://tartrerepub.blogspot.com/2018/08/cellcycle

5.1. Describe the importance of cell regeneration. (3)

5.2. In which phase does a cell spend most of its life? (1)

5.3. Explain how each cell is a duplicate of the original cell. (3)